Solution Requirements (Functional & Non-functional)

Date	25. September.2026	Team ID	SWTID-2026-4307
Project Name	AI Weather-Wise API	Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Form • Registration through Gmail • Registration through LinkedIn / OAuth
FR-2	User Confirmation & Authentication	• Confirmation via Email • Confirmation via OTP • JWT / API Key-based Authentication
FR-3	AI Microclimate Weather Forecasting	• Hyper-local temperature, humidity, and rainfall predictions • Real-time satellite & IoT sensor data fetching • Historical weather pattern analysis using Machine Learning
FR-4	Automated Risk Alerts & Recommendations	• Automated email & push notifications for sudden weather shifts • AI-driven agricultural & event planning action recommendations • Developer Dashboard for API key usage & analytics tracking

Non-functional Requirements

Following are the non-functional requirements of the proposed solution:

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The API documentation and dashboard should feature an intuitive, easy-to-navigate interface with clear endpoint examples for quick integration.
NFR-2	**Security**	Data transmission must be encrypted using SSL/TLS protocols, and user access controlled via secure API keys and token authentication.
NFR-3	**Reliability**	The system must guarantee 99.9% uptime with failover data pipelines to ensure uninterrupted weather forecasting services.
NFR-4	**Performance**	

NFR No.	Non-Functional Requirement	Description
		API response time should be under 200ms for standard weather queries, delivering low-latency real-time data under high query volumes.
NFR-5	**Availability**	The platform must remain accessible 24/7 across multiple cloud regions with zero planned downtime during critical forecast updates.
NFR-6	**Scalability**	The architecture must automatically scale horizontally to process thousands of concurrent API requests without performance degradation.